

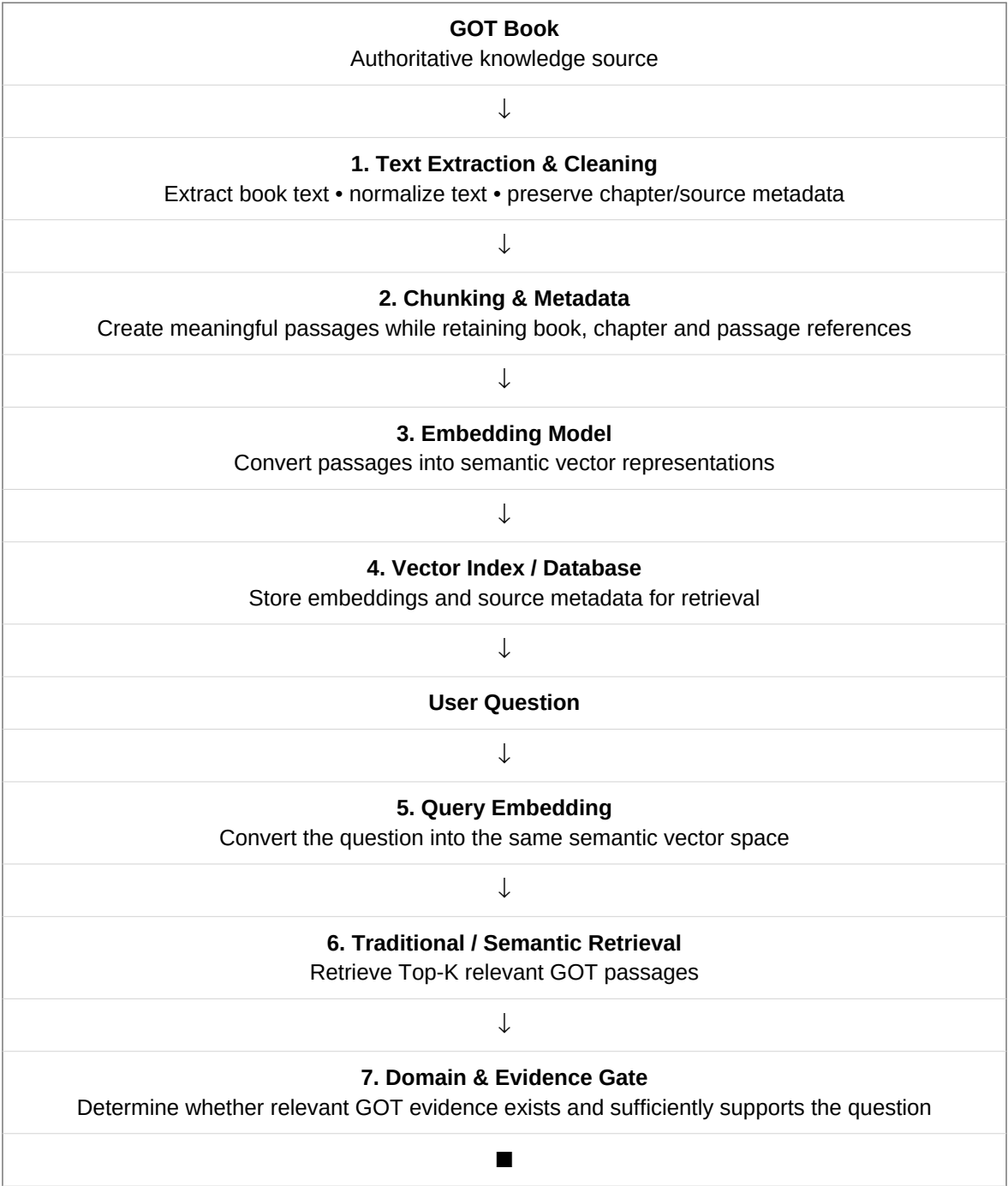
# Game of Thrones Book Q&A; Chatbot

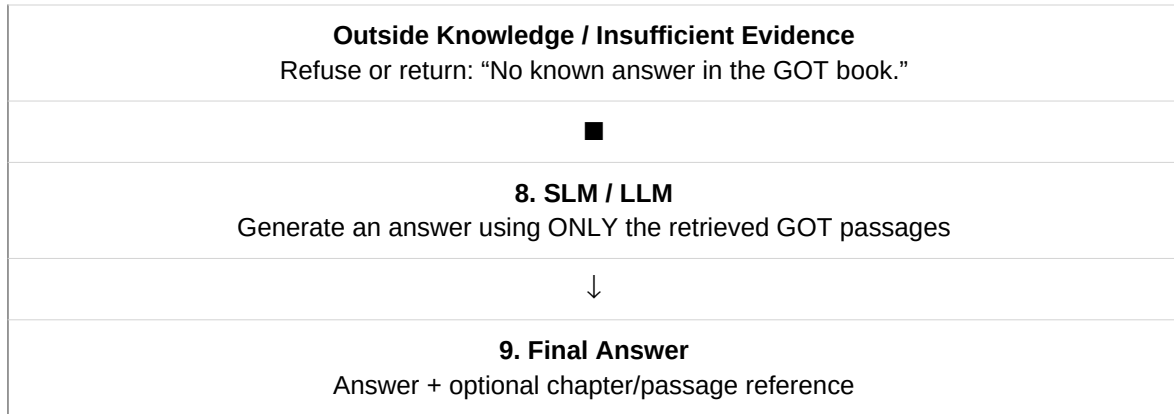
Final Architecture — Embedding + Traditional Retrieval + SLM/LLM

## 1. Objective

Build a question-answering chatbot using the **Game of Thrones (GOT) book content as the only knowledge source**. Users can ask about characters, events, relationships, plots, actions, and other information contained in the book. The system must refuse or return an unknown response when the requested information is outside the book knowledge base.

## 2. Final Architecture





### 3. Knowledge Boundary

The GOT book is the **only source of truth**. The pretrained SLM/LLM is not permitted to supply factual knowledge from its general training. Its role is limited to understanding the question and generating a natural-language answer from retrieved GOT evidence.

### 4. Expected Behaviour

| Question                              | Expected result                                                  |
|---------------------------------------|------------------------------------------------------------------|
| Who killed [a character]?             | Answer if the event is supported by retrieved GOT passages.      |
| Who married [a character]?            | Answer from the book evidence if the relationship is documented. |
| What happened during [an event]?      | Summarize the relevant retrieved passages.                       |
| Who is Harry Potter?                  | Refuse / "No known answer in the GOT book."                      |
| What happened in a book not included? | Refuse / outside the knowledge boundary.                         |

### 5. Strict Answering Flow

**Question** → **Retrieve** → **Validate evidence** → **Generate answer**. The evidence gate is essential: do not rely only on an LLM instruction such as "answer only from the book." If retrieval does not provide sufficiently strong evidence, the generation step is skipped.

### 6. Example

**User:** Who married Robb Stark?

**Retrieval:** Relevant GOT passages are retrieved.

**Evidence Gate:** Checks whether the passages actually support the relationship.

**SLM/LLM:** Generates the answer only from those passages.

**Final:** Answer with an optional chapter/passage reference.

### 7. Recommended Technology Pattern

Use a pretrained embedding model for semantic retrieval, a vector index/database for efficient Top-K search, and a small language model (SLM) or larger LLM purely as the answer-generation layer. This keeps the architecture modular: each component can be replaced independently.

## 8. Core Design Principle

**Retrieve first → validate evidence → generate second.**

This architecture provides a controlled, book-grounded Q&A; system while minimizing hallucination and preventing answers based on knowledge outside the GOT book.